Math 4400, Fall 2016 **Quiz 5** Name: ____________________

Calculators, cell phones, or notes are not allowed. Each problem is worth 5 pts.

(1) Show that $\mathbb{F}_{13}[\sqrt{10}]$ is not a field and find the inverse of $1+\sqrt{10}$ in $\mathbb{F}_{13}[\sqrt{10}]^*$.

$6^2 = 36 \cong_{13} 10$ so $N(6+\sqrt{10}) = 36-10 \cong_{13} 0$ i.e. $6+\sqrt{10}$ is not invertible (in fact $(6+\sqrt{10})\cdot(6-\sqrt{10}) \cong_{13} 0$).
$(1+\sqrt{10})^{-1} = (1-\sqrt{10})\cdot(-9)^{-1} = (1-\sqrt{10})\cdot(-3) = -3+3\sqrt{10}$. (Note that $(-3)\cdot(-9) = 27 \cong_{13} 1$).

(2) Define $\zeta(s)$ and show that $\zeta(s)$ converges if $s > 1$.

$\zeta(s) = \sum_{i=1}^{\infty} \frac{1}{i^s}$. By the integral test, $\zeta(s)$ converges if and only if the integral $\int_1^{\infty} \frac{1}{x^s} dx$ converges. For $s > 1$, this integral is computed by

$$\int_1^{\infty} \frac{1}{x^s} dx = \lim_{m \longrightarrow \infty} \left[\frac{-s+1}{x^{s-1}} \right]_1^m = (1-s) \lim_{m \longrightarrow \infty} (\frac{1}{m^{s-1}} - 1) = -(1-s).$$